

Here's a **comprehensive, industry-standard list of day-to-day SOPs (Standard Operating Procedures)** used across the IT industry for all kinds of applications and software development — from startups to large enterprises.

I've organized it so it's **practical, interview-ready, and real-world**, not textbook fluff.

✔ **Standard SOPs Used Daily in the IT Industry**

SOP Category	SOP Name	Purpose / What It Covers	Who Uses It
Project Initiation	Requirement Gathering SOP	Collect business, functional, and non-functional requirements	BA, PM, Tech Lead
	Feasibility Analysis SOP	Technical, financial, and timeline feasibility check	Architect, PM
	Scope Definition SOP	Define in-scope & out-of-scope items	PM
	Stakeholder Communication SOP	Identify stakeholders & communication plan	PM
Planning & Management	Project Planning SOP	Timeline, milestones, resources	PM
	Risk Management SOP	Identify, analyze, mitigate risks	PM, Lead
	Change Request SOP	Handle scope & requirement changes	PM
	Estimation SOP	Effort, cost, and time estimation	Lead, PM
Development Process	SDLC SOP	End-to-end software lifecycle	Entire Team
	Agile / Scrum SOP	Sprint planning, daily stand-ups, reviews	Dev, QA, PM
	Coding Standards SOP	Naming, formatting, best practices	Developers
	Code Review SOP	Peer review & quality enforcement	Developers

	Branching Strategy SOP	GitFlow, trunk-based, feature branches	DevOps, Dev
Version Control	Source Code Management SOP	Git usage, commits, merges	Developers
	Pull Request SOP	PR creation, approval process	Developers
	Release Tagging SOP	Versioning and tagging	DevOps
Testing & QA	Test Case Design SOP	Writing test cases & scenarios	QA
	Test Execution SOP	Manual & automated testing	QA
	Bug Reporting SOP	Defect lifecycle & severity	QA
	Regression Testing SOP	Ensure old features still work	QA
	UAT SOP	User acceptance testing process	QA, Client
Build & CI/CD	Build Management SOP	Compile & package applications	DevOps
	CI Pipeline SOP	Automated builds & tests	DevOps
	CD Pipeline SOP	Automated deployments	DevOps
	Artifact Management SOP	Store & version build artifacts	DevOps
Deployment	Deployment SOP	Production & non-prod deployment steps	DevOps
	Rollback SOP	Safe rollback during failure	DevOps
	Environment Management SOP	Dev, QA, UAT, Prod separation	DevOps
Security	Secure Coding SOP	Prevent vulnerabilities	Developers
	Authentication & Authorization SOP	Access control standards	Dev, Security
	Vulnerability Management SOP	Scan & fix security issues	Security Team

	Data Privacy SOP	GDPR, compliance handling	Security
Database & Data	Database Design SOP	Schema, normalization	DB Admin
	Migration SOP	DB version upgrades	DB Admin
	Backup & Restore SOP	Data protection	DB Admin
	Data Validation SOP	Data accuracy & integrity	Dev, QA
Performance	Performance Testing SOP	Load, stress, scalability tests	QA
	Optimization SOP	Improve speed & efficiency	Dev
	Monitoring SOP	Track performance metrics	DevOps
Operations & Support	Incident Management SOP	Handle production incidents	Support
	Root Cause Analysis SOP	Identify incident causes	Support, Dev
	SLA Management SOP	Response & resolution timelines	Support
	Ticket Handling SOP	Support tickets workflow	Support
Monitoring & Logging	Logging SOP	Application logs standardization	Dev
	Alerting SOP	System alerts & thresholds	DevOps
	Health Check SOP	Application availability checks	DevOps
Documentation	Technical Documentation SOP	Architecture & design docs	Dev
	API Documentation SOP	Swagger/OpenAPI standards	Dev
	User Manual SOP	End-user documentation	Tech Writer
Compliance & Governance	Audit SOP	Internal & external audits	Compliance
	Access Control SOP	Role-based access	IT Admin

	License Management SOP	Software license compliance	IT Admin
AI / Data / ML (Modern IT)	Data Collection SOP	Data sourcing & validation	Data Team
	Model Training SOP	ML model training steps	ML Engineer
	Model Validation SOP	Accuracy & bias checks	ML Engineer
	Model Deployment SOP	Serving ML models	MLOps
	AI Ethics SOP	Responsible AI usage	AI Team
Disaster Recovery	DR Planning SOP	Disaster preparedness	IT Ops
	Business Continuity SOP	Ensure minimal downtime	IT Ops
	Failover SOP	System redundancy	DevOps
HR & Team Operations	Onboarding SOP	New employee setup	HR, IT
	Offboarding SOP	Access removal	HR, IT
	Knowledge Transfer SOP	Handover process	Team
Client & Communication	Client Communication SOP	Status updates & reporting	PM
	Escalation SOP	Issue escalation hierarchy	PM
	Demo & Review SOP	Product demos & reviews	Dev, PM

□ How This Is Used in Real Companies

- **Startups:** Lightweight SOPs (Agile, Git, CI/CD)
- **Mid-size companies:** Formal SOPs for QA, security, DevOps
- **MNCs:** Strict SOPs + compliance, audits, SLAs, DR

❑ Interview Tip (Very Important)

If asked:

“Do you follow SOPs in your daily work?”

Say:

“Yes, we follow SOPs for SDLC, coding standards, code review, testing, CI/CD, deployment, incident handling, and documentation to ensure quality, security, and scalability.”

If you want next:

- ✓ **SOPs mapped to Fresher / Experienced roles**
- ✓ **SOPs specific to Java / Python / AI / DevOps**
- ✓ **Downloadable SOP checklist (PDF / Excel)**

Below is a **clear, practical mapping of SOPs to Fresher vs Experienced roles** that companies *actually* expect people to follow day-to-day. This is **interview-ready, resume-ready, and real-world accurate**.

✓ SOPs Mapped to Fresher vs Experienced Roles (IT Industry)

📌 **Fresher = Execution + Discipline**

📌 **Experienced = Ownership + Optimization + Decision-Making**

1 Project & SDLC SOPs

SOP	Fresher (0–2 yrs)	Experienced (3+ yrs)
SDLC SOP	Follow defined phases	Define & optimize SDLC
Requirement Understanding	Read & clarify tasks	Validate & refine requirements
Task Estimation	Estimate own tasks	Project/module estimation
Agile/Scrum SOP	Attend standups	Lead standups & planning
Change Request SOP	Follow changes	Analyze impact & approve

2 Coding & Development SOPs

SOP	Fresher	Experienced
Coding Standards SOP	Follow rules strictly	Define & enforce standards
Code Implementation	Write assigned modules	Design architecture
Secure Coding SOP	Avoid basic mistakes	Review & improve security
Code Optimization SOP	Basic refactoring	Performance tuning
Reusability SOP	Use existing code	Create reusable frameworks

3 Version Control & Git SOPs

SOP	Fresher	Experienced
Git Usage SOP	Clone, commit, push	Branching strategy
Pull Request SOP	Create PRs	Review & approve PRs
Merge SOP	Resolve simple conflicts	Handle complex merges
Release Tagging SOP	Follow tagging rules	Manage release versions

4 Testing & Quality SOPs

SOP	Fresher	Experienced
Unit Testing SOP	Write basic tests	Design test strategy
Bug Fix SOP	Fix assigned bugs	Root cause analysis
Regression SOP	Execute test cases	Plan regression cycles
UAT SOP	Support UAT	Coordinate with client

5 CI/CD & DevOps SOPs

SOP	Fresher	Experienced
CI Pipeline SOP	Trigger builds	Design pipelines
CD SOP	Deploy to non-prod	Production deployments
Environment SOP	Use environments	Manage environments
Rollback SOP	Follow rollback steps	Design rollback strategy

6 Database & Data SOPs

SOP	Fresher	Experienced
CRUD SOP	Write basic queries	Optimize complex queries
DB Migration SOP	Follow scripts	Design migrations
Backup SOP	Verify backups	Define backup policy
Data Integrity SOP	Validate data	Enforce consistency

7 Security & Compliance SOPs

SOP	Fresher	Experienced
Access Control SOP	Use assigned access	Define access roles
Secure Config SOP	Follow configs	Harden systems
Vulnerability SOP	Fix reported issues	Perform security reviews
Compliance SOP	Follow rules	Ensure audit readiness

8 Production Support & Operations SOPs

SOP	Fresher	Experienced
Incident SOP	Monitor & report	Lead incident resolution

Ticket Handling SOP	Resolve L1 tickets	Handle L2/L3 tickets
SLA SOP	Follow SLA timelines	Define & enforce SLAs
RCA SOP	Assist RCA	Lead RCA & prevention

9 Documentation & Communication SOPs

SOP	Fresher	Experienced
Technical Docs SOP	Update docs	Create architecture docs
API Docs SOP	Add endpoints	Design APIs
Client Communication	Internal updates	Client-facing discussions
Knowledge Transfer SOP	Learn & receive KT	Deliver KT sessions

☐ Leadership & Ownership SOPs

SOP	Fresher	Experienced
Decision SOP	Follow decisions	Make technical decisions
Mentorship SOP	Learn from seniors	Mentor juniors
Escalation SOP	Escalate issues	Handle escalations
Continuous Improvement	Learn tools	Drive improvements

☐ One-Line Summary (Very Interview-Friendly)

- **Fresher:**
"I follow SOPs related to coding, testing, Git, CI/CD, documentation, and incident reporting."
- **Experienced:**
"I own SOPs related to architecture, code reviews, deployments, security, performance, incident management, and mentoring."

❑ Resume Keywords (Copy-Paste Ready)

Fresher

SDLC Compliance, Coding Standards, Git Workflow, Unit Testing, Bug Fixing, CI/CD Execution, Documentation, Agile Participation

Experienced

SOP Ownership, Architecture Design, Code Review, CI/CD Design, Production Support, Security Compliance, RCA, Team Mentorship

If you want next:

- ✔ **Role-wise SOPs (Java / Python / QA / DevOps / AI)**
- ✔ **SOPs mapped to L1 / L2 / L3 support**
- ✔ **SOP-based interview Q&A**

Just say the word ❑

Nice — this is **exactly** how real IT support teams are structured ☑

Below is a **clear, industry-accurate mapping of SOPs to L1 / L2 / L3 support levels**, used in service desks, product companies, SaaS, BFSI, and MNC IT operations.

This is **interview-ready, process-ready, and day-to-day realistic**.

✔ SOPs Mapped to L1 / L2 / L3 Support (IT Industry)

☑ **L1 = First Line (Monitoring & Basic Resolution)**

☑ **L2 = Technical Analysis & Fix**

☑ **L3 = Expert / Engineering / Root Cause Owner**

1 Incident Management SOP

SOP	L1 Support	L2 Support	L3 Support
Incident Detection	Monitor alerts	Analyze incident	Validate system failure
Incident Logging	Create ticket	Update details	Provide technical notes
Incident Categorization	Basic category	Correct category	Final classification

Incident Resolution	Basic fixes	Code/config fixes	Permanent fix
Escalation SOP	Escalate to L2	Escalate to L3	Decide escalation
Closure SOP	Close simple tickets	Approve closure	Validate resolution

2 Ticket Handling SOP

SOP	L1	L2	L3
Ticket Creation	Create ticket	Review ticket	Validate issue
Priority Assignment	Follow matrix	Adjust priority	Approve priority
SLA Tracking	Track SLA	Prevent breach	Own SLA
Ticket Updates	User updates	Technical updates	Final solution
Ticket Closure	Close resolved	Approve closure	Audit closure

3 Monitoring & Alert SOP

SOP	L1	L2	L3
System Monitoring	24×7 monitoring	Analyze alerts	Define monitoring rules
Alert Acknowledgement	Acknowledge alerts	Investigate root	Improve alert logic
False Alert Handling	Report	Fix thresholds	Redesign monitoring

4 Application Support SOP

SOP	L1	L2	L3
App Issue Handling	Restart, clear cache	Config/app fix	Code-level fix
Log Analysis	Collect logs	Analyze logs	Enhance logging
Feature Support	User guidance	Functional support	Feature redesign
Performance Issues	Identify slowness	Tune configs	Architecture optimization

5 Database & Data SOP

SOP	L1	L2	L3
Data Issue Reporting	Identify issue	Query & fix	Data model redesign
Backup SOP	Verify backup status	Restore test	Define backup strategy
Data Integrity SOP	Report mismatch	Fix data	Prevent recurrence

6 Deployment & Release SOP

SOP	L1	L2	L3
Deployment Monitoring	Observe deployment	Fix deployment issues	Approve release
Rollback SOP	Trigger rollback	Execute rollback	Design rollback plan
Release Validation	Basic checks	Functional validation	Final sign-off

7 Security & Access SOP

SOP	L1	L2	L3
Access Requests	Create ticket	Grant access	Approve access
Security Incident	Report issue	Investigate	Final mitigation
Policy Compliance	Follow policies	Enforce policies	Define policies

8 Change Management SOP

SOP	L1	L2	L3
Change Request	Log CR	Impact analysis	Approve/reject
Change Implementation	Monitor	Implement change	Design change

Post-Change Review	Report outcome	Analyze issues	Improve process
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9 Root Cause Analysis (RCA) SOP

SOP	L1	L2	L3
RCA Data Collection	Collect logs	Analyze patterns	Identify root cause
RCA Document	Assist	Draft RCA	Finalize RCA
Preventive Actions	Follow instructions	Implement fixes	Design prevention

☐ Client & Communication SOP

SOP	L1	L2	L3
Client Updates	Status updates	Technical explanation	Strategic communication
Escalation Handling	Notify seniors	Manage escalations	Resolve escalations
Post-Incident Review	Attend PIR	Present findings	Lead PIR

☐ One-Line Interview Answers

L1 Support

"I follow SOPs for incident logging, monitoring alerts, basic troubleshooting, SLA tracking, and escalation."

L2 Support

"I handle SOPs for deep technical troubleshooting, configuration fixes, deployment issues, and preventing SLA breaches."

L3 Support

"I own SOPs for root cause analysis, permanent fixes, architecture-level changes, and process improvements."

☐ Resume Keywords (Copy-Paste)

L1

Incident Monitoring, Ticket Logging, SLA Tracking,
First-Level Support, Alert Handling

L2

Technical Troubleshooting, Log Analysis,
Configuration Fixes, Deployment Support

L3

Root Cause Analysis, Production Fixes,
Architecture Improvements, Incident Ownership

If you want next (very popular requests):

- ✓ L1/L2/L3 real interview questions & answers
- ✓ ITIL-based SOP mapping
- ✓ Support SOP flow diagram (Incident → RCA → Closure)

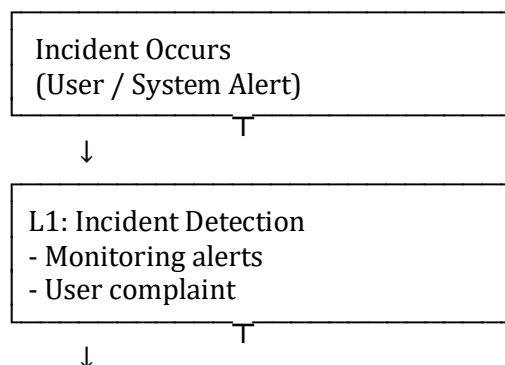
Below is a **clear, industry-standard Support SOP flow diagram** showing the **complete lifecycle from Incident → RCA → Closure**, exactly how it works in real IT environments (ITIL-aligned).

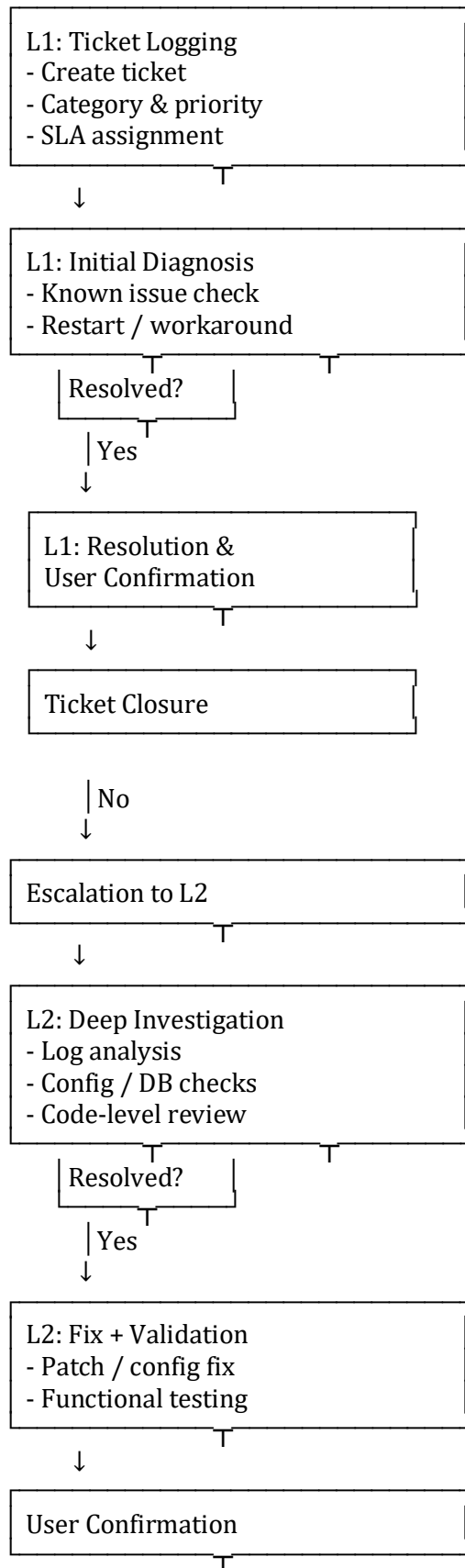
I'll give you:

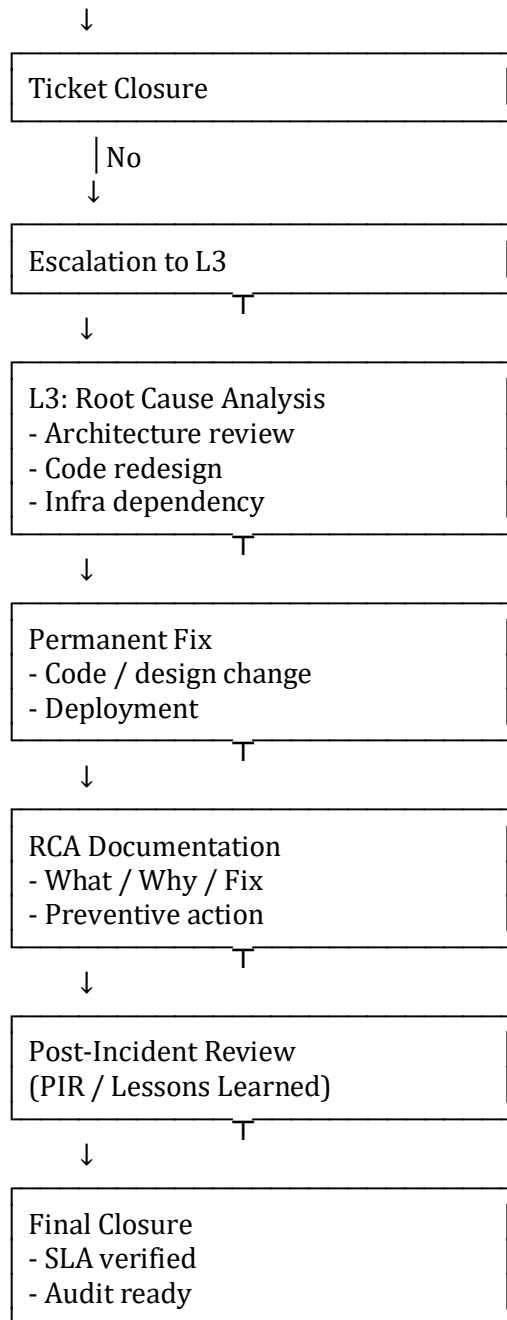
1. Visual flow (text-based diagram)
2. L1 / L2 / L3 responsibility mapping
3. Short interview-ready explanation

✓ Support SOP Flow Diagram

(Incident → RCA → Closure)







✓ L1 / L2 / L3 Responsibility Mapping (Quick View)

Stage	L1	L2	L3
Incident Detection	✓	✗	✗
Ticket Creation	✓	✗	✗

Initial Fix	✓	✗	✗
Deep Troubleshooting	✗	✓	✗
Configuration / Patch	✗	✓	✗
Root Cause Analysis	✗	△ Assist	✓
Permanent Fix	✗	△ Assist	✓
RCA Document	✗	Draft	Finalize
Closure Approval	✓ (simple)	✓	✓ (major)

✓ Interview-Ready Explanation (Perfect Answer)

“When an incident occurs, L1 handles detection, ticket logging, basic troubleshooting, and escalation.

L2 performs deep technical analysis, applies configuration or application-level fixes, and validates resolution.

If unresolved, L3 performs root cause analysis, delivers permanent fixes, documents RCA, and drives preventive improvements before final closure.”

✓ Key ITIL Terms You Can Drop (Big Impact)

- Incident Management
- SLA Tracking
- Escalation Matrix
- Root Cause Analysis (RCA)
- Post-Incident Review (PIR)
- Corrective & Preventive Actions (CAPA)

If you want next:

- ✓ Incident priority matrix (P1-P4)
- ✓ Real RCA document template
- ✓ Support SOP flow as PNG / PPT / PDF